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VOICE COMMAND RECOGNITION SYSTEM

Abstract

Disclosed herein are system, apparatus, article of manufacture, method and/or computer program product embodiments, and/or combinations and sub-combinations thereof, for a voice command recognition system (VCR). An example embodiment operates by receiving a voice command directed to controlling a device, the voice command including a wake command and an action command. An amplitude of the wake command is determined. A gain adjustment for the voice command is calculated based on a comparison of the amplitude of the wake command to a target amplitude. An amplitude of the action command is adjusted based on the calculated gain adjustment for the voice command based on the comparison of the amplitude of the wake command to the target amplitude. A device command for controlling the device is identified based on the action command comprising the adjusted amplitude. The device command is provided to the device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/315,704, filed on May 10, 2021, entitled “VOICE COMMAND RECOGNITION SYSTEM,” which is incorporated herein by reference in its entirety.

FIELD

[0002] This disclosure is generally directed to recognizing voice commands for voice responsive devices.

BACKGROUND

[0003] A voice responsive device (VRD) is an electronic device that responds to audible or voice commands spoken by users. Examples of VRDs include digital assistants, televisions, media devices, smart phones, computers, tablets, appliances, smart televisions, and internet of things (IOT) devices, to name just some examples.

[0004] One of the challenges with processing voice commands is that these commands are often spoken and thus received at different volumes. Sometimes a user will speak loudly, other times softly, and different users speak differently. As such, it can become challenging to recognize and process voice commands with such wide variances in audio quality or volume.

SUMMARY

[0005] Provided herein are system, apparatus, article of manufacture, medium, method and/or computer program product embodiments, and/or combinations and sub-combinations thereof, for a voice command recognition system (VCR).

[0006] An embodiment include a method. The method may include receiving a voice command directed to controlling a device, the voice command including a wake command and an action command. An amplitude of the wake command is determined. A gain adjustment for the voice command is calculated based on a comparison of the amplitude of the wake command to a target amplitude. An amplitude of the action command is adjusted based on the calculated gain adjustment for the voice command based on the comparison of the amplitude of the wake command to the target amplitude. A device command for controlling the device is identified based on the action command comprising the adjusted amplitude. The device command is provided to the device.

[0007] Another embodiment includes a system that may include a memory and at least one processor communicatively coupled to the memory. The processor may be configured to receive a voice command directed to controlling a device, the voice command including a wake command and an action command. An amplitude of the wake command is determined. A gain adjustment for the voice command is calculated based on a comparison of the amplitude of the wake command to a target amplitude. An amplitude of the action command is adjusted based on the calculated gain adjustment for the voice command based on the comparison of the amplitude of the wake command to the target amplitude. A device command for controlling the device is identified based on the action command comprising the adjusted amplitude. The device command is provided to the

device.

[0008] A further embodiment includes a non-transitory computer-readable medium having instructions stored thereon that, when executed by at least one computing device, causes the computing device to perform operations. These operations may include receiving a voice command directed to controlling a device, the voice command including a wake command and an action command. An amplitude of the wake command is determined. A gain adjustment for the voice command is calculated based on a comparison of the amplitude of the wake command to a target amplitude. An amplitude of the action command is adjusted based on the calculated gain adjustment for the voice command based on the comparison of the amplitude of the wake command to the target amplitude. A device command for controlling the device is identified based on the action command comprising the adjusted amplitude. The device command is provided to the device.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0009] The accompanying drawings are incorporated herein and form a part of the specification.

[0010] FIG. 1 illustrates a block diagram of a voice command recognition system (VCR), according to some embodiments.

[0011] FIG. 2 illustrates a block diagram of a multimedia environment, according to some embodiments.

[0012] FIG. 3 illustrates a block diagram of an example media device, according to some embodiments.

[0013] FIG. 4 is a flowchart illustrating example operations for providing a voice command recognition system (VCR), according to some embodiments.

[0014] FIG. 5 is a flowchart illustrating example operations for providing a voice command recognition system (VCR) with beep suppression, according to some embodiments.

[0015] FIG. 6 illustrates an example computer system useful for implementing various embodiments.

[0016] In the drawings, like reference numbers generally indicate identical or similar elements. Additionally, generally, the left-most digit(s) of a reference number identifies the drawing in which the reference number first appears.

DETAILED DESCRIPTION

[0017] Provided herein are system, apparatus, device, method, medium, and/or computer program product embodiments, and/or combinations and sub-combinations thereof, for recognizing voice commands.

[0018] A voice responsive device (VRD) is an electronic device that responds to audible or voice commands (also called audio commands) spoken by users. Examples of VRDs include digital assistants, televisions, media devices, smart phones, computers, tablets, appliances, smart televisions, and internet of things (IOT) devices, to name just some examples.

[0019] One of the challenges with processing voice commands is that these commands are often spoken and thus received at different volumes and from varying distances relative to a receiving microphone. Sometimes a user will speak loudly, other times softly, and different users speak differently. Users may also be standing near the microphone or across the room far away from it, which can produce variances in the quality and loudness of the sounds detected or received by the microphone. As such, it can become challenging to recognize and process audio commands with such wide variances in audio quality or volume.

[0020] FIG. 1 illustrates a block diagram of a voice command recognition system (VCR) 102, according to some embodiments. Voice command recognition system (VCR) 102 may recognize

and process an audible or voice command **104** for operating a voice responsive device (VRD) **106**. In some embodiments, VCR **102** may normalize or smooth out the audio characteristics of a voice command **104** for improved audio processing.

[0021] For example, in some embodiments, VCR **102** may adjust the amplitude of a voice command **104** based on variances between the received characteristics of the voice command **104** relative to target characteristics for processing the voice command **104**. These audio adjustments may enable VCR **102** and/or other audio input/output systems to more quickly and/or accurately process the adjusted voice command (relative to the received non-adjusted command). The audio processing may include, for example, identifying a device command **108**, converting text-to-speech, performing language translations, or performing other audio processing functionality.

[0022] In an embodiment, a user **110** may speak into a microphone (mic) **112** to operate VRD **106**. Mic **112** may be configured to record or receive audio from a user **110** and/or the environment of the user **110**, which may include background noises. Mic **112** may provide the received audio (e.g., speech, noise, and/or other audio) to VCR **102**. VCR **102** may receive the voice command **104**, identify different sub-commands or speech portions within the received audio, adjust audio characteristics, and identify a device command **108** for operating a VRD **106**.

[0023] In an embodiment, a voice command **104** may include different portions or sub-commands, such as a wake command **114** (also called a wake word) and an action command **116**. Wake command **114** may be one or more initial words that signal that a VRD action or operational command **116** is to follow. For example, for the AMAZON ECHO (e.g., VRD **106**), a voice command **104** may be “ALEXA, turn on the family room lights,” of which “ALEXA” may be the wake command **114** and “turn on the family room lights” may be the action **116** command.

[0024] VRD **106** may be an electronic device that responds to audible or voice commands **104** spoken by a user **110**. Examples of a VRD **106** include, but are not limited to, digital assistants, televisions, media devices, smart phones, computers, tablets, appliances, and internet of things (IOT) devices, to name just some examples.

[0025] In an embodiment, VRD **106** may be connected to one or more other devices (e.g., sound system, speaker, light, television, thermostat, home security, computer, IoT device), which are operational through voice commands **104** issued to VRD **106**. For example, the AMAZON ECHO (e.g., VRD **106**) may be connected to a sound bar which may be controlled through the AMAZON ECHO (e.g., turn on/off, volume up/down, increase bass, adjust treble, change audio environment setting, etc.).

[0026] One of the challenges with processing voice commands **104** is that the audio received from a user **110** can vary in its volume, quality, or other audio characteristics. A user **110** may speak in a loud or soft voice depending on their mood or other environmental circumstances. For example, the user **110** may increase their voice while speaking if they are excited or there is background noise (such as a police siren), or decrease their voice if their baby is sleeping or they are tired. Such volume changes or other alterations in the user **110**'s voice while speaking the voice command **104** may make it difficult to accurately process the user's speech or voice and recognize the action command **116** in the voice command **104**.

[0027] In an embodiment, a gain processor **118** may normalize or apply audio characteristic adjustments to voice command **104** to aid in speech or audio processing. In an embodiment, gain processor **118** may apply a gain adjustment **120** to increase (or decrease) the loudness of at least a portion of the voice command **104**. While gain is identified as an exemplary audio characteristic that is adjusted by VCR **102**, it is understood that in other embodiments, other audio characteristics of voice command **104** (such as bass, treble, pitch, speed, etc.) may be adjusted in a similar manner as described herein, and the audio adjustments are not limited to gain, but may include any audio characteristic(s).

[0028] In an embodiment, VCR **102** may apply audio adjustments to the action command **116** portion of voice command **104** based on an audio analysis of the wake command **114** portion of

voice command **104**. In an embodiment, gain processor **118** may calculate, measure, or determine a wake amplitude **122** of wake command **114**. In an embodiment, amplitude may be measure of a strength or level of energy or sound pressure.

[0029] Gain processor **118** may compare wake amp **122** to a target amplitude **124**. Target amp **124** may be an ideal, goal, or minimum measure of amplitude for audio that is to undergo voice or speech processing. In an embodiment, the closer an amplitude of audio is to target amp **124**, the more accurate audio processing may be performed on the audio.

[0030] Gain processor **118** may calculate a gain adjustment **120**. In an embodiment, the gain adjustment **120** may be a difference between wake amp **122** and target amp **124**. For example, if wake amp **122** is -30 decibels (dB) and target amplitude is -25 dB, the gain adjustment **120** may be 5 dB.

[0031] In an embodiment, gain processor **118** may apply the gain adjustment **120** to an action amplitude **126** (e.g., amplitude of action command **116**) to generate a gained action command **128**. In an embodiment, gain processor **118** may use just-in-time computing processes to apply gain adjustment **120** to action command **116**, such that the audio adjustment is performed in real-time without first computing or identifying action amp **126** of action command **116**. This real-time application of gain adjustment **120** to action command **116** may save computing resources and time, reducing latency while also increasing the accuracy of speech processing on gained action command **128**.

[0032] In an embodiment, VCR **102** may then identify a device command **108** for operating VRD **106** from the gained action command **128**. In an embodiment, identifying device command **108** may include converting gained action command **128** to text, and comparing the text to a set of valid operational commands for VRD **106**.

[0033] FIG. 1 illustrates one example embodiment and configuration of audio processing by VCR **102**. However, in other embodiments, the functionality and processing of VCR **102** as described herein may be organized differently. For example, mic **112** and/or VCR **102** functionality may be integrated within VRD **106** as a single device. Or, for example, mic **112** and/or VCR **102** functionality may be integrated into a mobile device or remote control associated or communicatively coupled to VRD **106**). Or, for example, VCR **102** functionality may exist on a cloud network that communicates with the remote control and/or VRD **106** communicatively coupled to mic **112**.

[0034] In an embodiment, VCR **102** may track historical data **130** which may be used to apply a correction **132** to the audio adjustments of gain processor **118**. Historical data **130** may include a history of the various gain adjustments **120** applied to voice commands **104** by VCR **102**. In an embodiment, the historical data **130** may be organized based on a user or remote control or other device from which the voice command **104** is received.

[0035] In some embodiments, VCR **102** may also be configured to differentiate between different users **110** using one or more speaker detection processes. For example, based on a cadence or timbre of the voice or speech of user **110**, the VCR **102** may use one or more machine learning algorithms to classify and identify the user **110**.

[0036] In some embodiments, VCR **102** may be configured to differentiate between different users **110** using data from one or more sensors in a remote control that a user **110** may have handled or may be handling. For example, based on a velocity or speed at which the user **110** picks up the remote control or a movement pattern of the remote control, VCR **102** may use one or more machine learning algorithms to classify and identify the user **110**.

[0037] For example, the one or more machine learning algorithms may include classification algorithms, such as, but not limited to, k-Nearest Neighbors, decision trees, naive bayes, random forest, or gradient boosting. Additionally, the one or more machine learning algorithms may be used to detect certain velocities and/or movement patterns and associate such velocities and/or movement patterns with a given user. In an embodiment, historical data **130** may include

movement patterns corresponding with different users **110**. As such, when the velocities and/or movement patterns are detected again in the future, VCR **102** may identify a particular user **110** accordingly.

[0038] The machine learning algorithms may further include an association algorithm, such as, but not limited to, an apriori algorithm, eclat algorithm, or a frequent-pattern growth (FP-growth) algorithm. Based on identifying which user **110** is interacting with the remote control, the VCR **102** may modify a time period for evaluating the RMS amplitude of the voice command versus time. For example, VCR **102** may increase or decrease the time period for evaluating the voice command for each individual user based on the cadence and/or timbre. For example, for some users that speak with a slower cadence than other users, VCR **102** may increase the time period for evaluating the voice command, whereas, for some users that speak with a faster cadence, VCR **102** may decrease the time period for evaluating the voice command **104**.

[0039] As noted above, the adjustments (such as gain adjustment **120** or cadence or other audio adjustments) may be tracked in a database or other storage as historical data **130**, which may be organized on a per user basis. For example, historical data **130** may include a first set of data for audio adjustments applied to voice commands **104** from user A, and a second set of data for audio adjustments applied to voice commands **104** from user B. In some embodiments, historical data **130** may be stored locally at VRD **106** or VCR **102**, or accessed from the cloud or via other network capabilities.

[0040] In an embodiment, VCR **102** may calculate a correction **132** for user A. VCR **102**, or another machine learning or cloud-based process, may compare the historical audio adjustments for user A against target amp **124**. And if the gained action command **128** for user A is statistically (average, mean, median, etc.) 6 dBs below target amp **124**, then correction **132** may be 6 dBs. Then, for example, when a subsequent voice command **104** is identified as being received from user A, gain processor **118** may adjust the action command **116** by applying both gain adjustment **120** and correction **132**. However, a subsequent command from user B would not include correction **132** for user A. Or, for example, correction **132** may be calculated across multiple or all users **110** of the system.

[0041] In some embodiments, historical data **130** may be used to identify and/or discard (or not record in historical data **130**) outliers of voice commands **104**. For example, if a user **110** screams into the mic **112**, this may register an outlier relative to normal spoken voice commands **104** from user **110**. Less or no gain may be applied to a screamed/loud voice command **104**. Similarly, if the user **110** is speaking from across the room, this may require a large outlier in terms of the gain adjustment **120** applied. Both of these outliers may be identified based on comparing the gain adjustment **120** to a median gain adjustment from historical data **130**, and may not be used to calculate correction **132**.

[0042] As noted above, the VRD **106** may be voice responsive and operable to recognize and process voice commands **104** spoken by users **110**. A voice command **104** typically includes a wake command **114** followed by an action command **116**. In some embodiments, an audio receiver **134** of VCR **102** may output an audible notification or beep **136** indicating the voice command **104** was received or that the VRD **106** is listening for an action command **116**. In an embodiment, beep **136** may be an audible notification or sound that is output through a speaker of VRD **106**, a remote control, a mobile device, or another device with audio output capabilities.

[0043] In an embodiment, the beep **136** may be output upon identification of the wake command **114**, indicating the VRD **106** is ready to receive an action command **116**. However, some users **110** may speak fast or continuously, and as such, if user **110** speaks the action command **116** immediately after the wake command **114**, without waiting for beep **136** before speaking the action command **116**, then the beep **136** may be received by mic **112** at the same time as the action command **116**, and thus interfere with the ability to accurately recognize the action command **116** (because beep **136** may be received and interpreted as part of voice command **104**).

[0044] Accordingly, in some embodiments, audio receiver **134** may be configured to detect whether the voice command **104** is a continuous stream of speech such that the user **110** does not pause for any significant length of time (e.g., at least long enough for beep **136** to be output) between speaking the wake command **114** and speaking the action command **116**.

[0045] To determine whether the voice command is a continuous stream of speech, audio receiver **134** may analyze the voice command to detect bursts of energy in a specified period of time. For example, audio receiver **134** may analyze an average root mean square (RMS) amplitude of the voice command **104** versus time, and the audio receiver **134** may determine that the voice command **104** is a continuous stream of speech based on such analysis. For example, in some embodiments, audio receiver **134** may analyze a ratio of a peak-to-average amplitude of successive frames of the voice command using an exponential-decay smoothing. As an example, the audio receiver **134** may use a sampling rate of 16 k/sec and each frame may include 256 samples. In some embodiments, a ratio greater than or equal to one (1) may indicate that continuous speech is present, whereas a ratio less than one (1) may indicate that continuous speech is not present.

[0046] Based on whether the audio receiver **134** detected the continuous stream of speech, the audio receiver **134** may a remote control **210** or VRD **106** to selectively provide or not provide beep **136**. For example, if a continuous stream of speech is detected, audio receiver **134** may lower the volume on, mute, or otherwise suppress the beep **136**. In another embodiment, audio receiver **134** may not send a beep **136** signal for audio output. Or, for example, if the continuous stream of speech is not detected, VRD **106** may automatically output beep **136** as it was configured to do, or audio receiver **134** may provide the beep **136** signal for output. In this way, audio receiver **134** ensures that the audible notification (i.e., beep **136**) does not interfere with recognizing the action command **116** following the wake command **114** in the spoken voice command **104** when the user **110** does not pause between speaking the wake command **114** and speaking the action command **116**.

[0047] In some embodiments, in addition to suppressing the audible notification when the voice command is determined to be a continuous stream of speech, the audio receiver **134** may cause the remote control and/or the VRD **106** to provide an alternative non-audio notification to the user **110** responsive to detecting the wake command **114**. For example, a visual or haptic notification may be provided to the user **110** via the remote control (using any well-known haptic module in the remote control), and/or the VRD **106**. In this way, the user **110** may still be notified that VRD **106** or remote control received and/or recognized the wake command **114** and is processing the voice command **104** or awaiting the action command **116**.

[0048] Solely for purposes of convenience, and not limitation, embodiments of this disclosure are described with respect to an example multimedia environment **202** shown in FIG. **2**. However, this example application is provided solely for illustrative purposes, and is not limiting. Embodiments of this disclosure are applicable to any VRD in any application and/or environment, as will be understood by persons skilled in the relevant art(s) at least based on the teachings contained herein.

[0049] FIG. **2** illustrates a block diagram of a multimedia environment **202**, according to some embodiments. In a non-limiting example, multimedia environment **202** may be directed to streaming media. However, this disclosure is applicable to any type of media (instead of or in addition to streaming media), as well as any mechanism, means, protocol, method and/or process for distributing media.

[0050] The multimedia environment **202** may include one or more media systems **204**. A media system **204** could represent a family room, a kitchen, a backyard, a home theater, a school classroom, a library, a car, a boat, a bus, a plane, a movie theater, a stadium, an auditorium, a park, a bar, a restaurant, or any other location or space where it is desired to receive and play streaming content. User(s) **232** may operate with the media system **204** to select and consume content.

[0051] Each media system **204** may include one or more media devices **206** each coupled to one or more display devices **208**. It is noted that terms such as “coupled,” “connected to,” “attached,”

“linked,” “combined” and similar terms may refer to physical, electrical, magnetic, logical, etc., connections, unless otherwise specified herein.

[0052] Media device **206** may be a streaming media device, DVD or BLU-RAY device, audio/video playback device, cable box, and/or digital video recording device, to name just a few examples.

[0053] In embodiments of this disclosure, media device **206** may be a voice responsive device (VRD) that responds to voice commands spoken by users **232**.

[0054] Display device **208** may be a monitor, television (TV), computer, smart phone, tablet, wearable (such as a watch or glasses), appliance, internet of things (IoT) device, and/or projector, to name just a few examples. In some embodiments, media device **206** can be a part of, integrated with, operatively coupled to, and/or connected to its respective display device **208**.

[0055] Each media device **206** may be configured to communicate with network **218** via a communication device **214**. The communication device **214** may include, for example, a cable modem or satellite TV transceiver. The media device **206** may communicate with the communication device **214** over a link **216**, wherein the link **216** may include wireless (such as WiFi) and/or wired connections.

[0056] In various embodiments, the network **218** can include, without limitation, wired and/or wireless intranet, extranet, Internet, cellular, Bluetooth, infrared, and/or any other short range, long range, local, regional, global communications mechanism, means, approach, protocol and/or network, as well as any combination(s) thereof.

[0057] Media system **204** may include a remote control **210**. The remote control **210** can be any component, part, apparatus and/or method for controlling the media device **206** and/or display device **208**, such as a remote control, a tablet, laptop computer, smartphone, wearable, on-screen controls, integrated control buttons, audio controls, or any combination thereof, to name just a few examples. In an embodiment, the remote control **210** wirelessly communicates with the media device **206** and/or display device **208** using cellular, Bluetooth, infrared, etc., or any combination thereof.

[0058] The remote control **210** may include a microphone **212** for receiving voice commands from users **232**. Data representing such voice commands may be transmitted to the media device **206** using any wireless means, such as RF, WiFi, cellular, infrared, etc. Also or alternatively, the media device **206** may include a microphone (not shown) for receiving voice commands from users **232**.

[0059] The remote control **210** may also include a speaker **213** and a sensor **215**, e.g., an accelerometer or a gyroscope, which are further described below.

[0060] The multimedia environment **202** may include a plurality of content servers **220** (also called content providers or sources **220**). Although only one content server **220** is shown in FIG. 2, in practice the multimedia environment **202** may include any number of content servers **220**. Each content server **220** may be configured to communicate with network **218**.

[0061] Each content server **220** may store content **222** and metadata **224**. Content **222** may include any combination of music, videos, movies, TV programs, multimedia, images, still pictures, text, graphics, gaming applications, advertisements, programming content, public service content, government content, local community content, software, and/or any other content or data objects in electronic form.

[0062] In some embodiments, metadata **224** comprises data about content **222**. For example, metadata **224** may include associated or ancillary information indicating or related to writer, director, producer, composer, artist, actor, summary, chapters, production, history, year, trailers, alternate versions, related content, applications, and/or any other information pertaining or relating to the content **222**. Metadata **224** may also or alternatively include links to any such information pertaining or relating to the content **222**. Metadata **224** may also or alternatively include one or more indexes of content **222**, such as but not limited to a trick mode index.

[0063] The multimedia environment **202** may include one or more system servers **226**. The system

servers **226** may operate to support the media devices **206** from the cloud. It is noted that the structural and functional aspects of the system servers **226** may wholly or partially exist in the same or different ones of the system servers **226**.

[0064] The media devices **206** may exist in thousands or millions of media systems **204**.

Accordingly, the media devices **206** may lend themselves to crowdsourcing embodiments and, thus, the system servers **226** may include one or more crowdsource servers **228**.

[0065] For example, using information received from the media devices **206** in the thousands and millions of media systems **204**, the crowdsource server(s) **228** may identify similarities and overlaps between closed captioning requests issued by different users **232** watching a particular movie. Based on such information, the crowdsource server(s) **228** may determine that turning closed captioning on may enhance users' viewing experience at particular portions of the movie (for example, when the soundtrack of the movie is difficult to hear), and turning closed captioning off may enhance users' viewing experience at other portions of the movie (for example, when displaying closed captioning obstructs critical visual aspects of the movie). Accordingly, the crowdsource server(s) **228** may operate to cause closed captioning to be automatically turned on and/or off during future streamings of the movie. This crowdsourcing example is described, for example, in U.S. Pat. No. 9,749,700 filed Nov. 21, 2016 and titled "Automatic Display of Closed Captioning Information."

[0066] The system servers **226** may also include an audio command processing module **230**. As noted above, the remote control **210** may include the microphone **212** and the speaker **213**. The microphone **212** may receive audio data from users **232** (as well as other sources, such as the display device **208**). The speaker **213** may provide audible notifications (such as beeps) to the user **232**. As noted above, in some embodiments, the media device **206** may be voice responsive, and the audio data may represent voice commands from the user **232** to control the media device **206** as well as other components in the media system **204**, such as the display device **208**.

[0067] In some embodiments, the audio data received by the microphone **212** in the remote control **210** is transferred to the media device **206**, which is then forwarded to the audio command processing module **230** in the system servers **226**. The audio command processing module **230** may operate to process and analyze the received audio data to recognize the user **232**'s voice command. The audio command processing module **230** may then forward the voice command back to the media device **206** for processing.

[0068] In some embodiments, the audio data may be alternatively or additionally processed and analyzed by an audio command processing module **316** in the media device **206** (see FIG. 3). The media device **206** and the system servers **226** may then cooperate to pick one of the voice commands to process (either the voice command recognized by the audio command processing module **230** in the system servers **226**, or the voice command recognized by the audio command processing module **216** in the media device **206**).). An example of such operation is described in U.S. application Ser. No. 16/032,868 filed Jul. 11, 2018 and titled "Local and Cloud Speech Recognition" (Atty. Docket No. 3634.1060001).

[0069] In some embodiments, one or more of the functionalities as described above with respect to VCR **102** may be performed by the audio command processing module **230** in system servers **226** as illustrated in FIG. 2. For example, audio command processing module **230** may perform an automatic gain adjustment of the voice command **104** to better ensure that the action command **116** in the voice command **104** is accurately recognized. To achieve this, VCR **102** may analyze an amplitude of the wake word or wake command **114** in the voice command **104**, and determine the nature of an automatic gain adjustment **120** to be applied to the wake command **114** so as to satisfy a minimum amplitude requirement associated with the speech to text conversion process as may be indicated by target amp **124**.

[0070] For example, the audio command processing module **230** may measure a difference between the amplitude of the wake command **114** and a target amplitude **124** to determine the

automatic gain adjustment **120** needed. In an embodiment, target amplitude **124** may include a range of acceptable values.

[0071] In some embodiments, the audio command processing module **230** may predict that the audio characteristics, such as loudness or gain, of the action command **116** portion of the voice command **104** may be at a same amplitude as the wake command **114**, and as such, apply the automatic gain adjustment **120** calculated based on the wake command **114** to the action command **116** to reach the target amplitude **124**. In this way, the audio command processing module **230** may proactively apply the automatic gain adjustment **120** to the action command **116**, rather than waiting to analyze the action command **116** itself, which could result in extra delays. As a result, the audio command processing module **230** provides for reduced processing time for converting the voice command **104** from speech to text.

[0072] In some embodiments, the audio command processing module **230** may also use historical information or data **130** of previous voice commands to determine the automatic gain adjustment **120**. For example, the audio command processing module **230** may analyze automatic gain adjustments applied to previous voice commands and predict what automatic gain adjustment should be for a current or subsequent voice command **104** based on the automatic gain adjustment applied to previous voice commands **104**. In an embodiment, this predicted gain may be applied to subsequent wake commands **114** and/or action commands **116**.

[0073] Furthermore, in some embodiments, the audio command processing module **230** may analyze historical data **130** to remove any anomalies. For example, the audio command processing module **230** may compare the amplitude of the wake commands **114** in the historical data **130** and identify instances where the wake amplitude **122** of a current wake command **114** is higher or lower in comparison to other wake amps **122** in historical data **130**.

[0074] For example, in some embodiments, the audio command processing module **230** may remove or separately store the anomalies using a median filter to smooth out the historical information or historical data **130**. As an example, the audio command processing module **230** may determine that an amplitude of the wake words may generally fall within a range, e.g., -18 dB to -22 dB, and the audio command processing module **230** may determine that any wake words with wake amplitudes **122** falling outside that range are anomalies and should be discarded from the historical data **130**. By removing any outliers from historical data **130**, the audio command processing module **230** may prevent applying too much or too little gain to the present or a subsequent action command **116** in a voice command **104** when using historical data **130**.

[0075] The audio command processing module **230** may also apply the automatic gain adjustment **120** in response to detecting that the voice command is a continuous stream of speech, as discussed above. In this way, the audio command processing module **230** may ensure that the automatic gain adjustment is being applied to actual voice commands spoken by users **232**, rather than other ambient noise.

[0076] In an embodiment, the audio command processing module **230** may include a feedback loop to improve the automatic gain adjustment **120**. For example, the audio command processing module **230** may analyze the amplitudes of both the wake word and the command in voice commands, and determine that the amplitude of the command is historically lower (or higher) than the amplitude of the wake word. Using this information, the audio command processing module **230** may determine a difference in the amplitudes of both the wake word or wake command **114** and the action command **116**, scale the automatic gain adjustment **120** or compute a correction **132** for the action command **116** so as to compensate for the difference in the amplitudes of the wake word and the action command **116**.

[0077] The feedback loop may also include an analysis of the amplitude of the voice command after the automatic gain adjustment to determine an accuracy of the automatic gain adjustment. For example, the analysis may include comparing a post-adjustment amplitude of the voice command to the target amplitude. Using this information, the audio command processing module **230** may

determine a correction factor **132** for adjusting the automatic gain adjustment applied to the action command. For example, the audio command processing module **230** may determine a difference between the post-adjustment amplitude and the target amplitude and compare this difference to a threshold value. When the difference exceeds the threshold value, the correction factor **132** may indicate that the automatic gain adjustment should be reduced (or increased) accordingly.

[0078] As discussed above, the audio command processing module **230** may be configured to differentiate between different users **232** using one or more speaker detection processes. Based on which user **232** is interacting with the remote control **210**, the audio command processing module **230** may use the historical information of that particular user to determine the gain to be applied to the command portion of the voice command. In some embodiments, when the audio command processing module **230** does not recognize a particular user **232**, the audio command processing module **230** may apply a gain adjustment to the command based on any of the embodiments discussed herein.

[0079] In some embodiments, the audio command processing module **230** may analyze a given voice command to determine whether it is an actual voice command issued by a user **232**. For example, the audio command processing module **230** may determine whether the voice command is an actual voice command based on whether content of the voice command matches an intent from among a plurality of intents that may be executed by the media device **206**. In the event that the content of the voice command does not match an intent from among the plurality of intents, the audio command processing module **230** may remove the voice command from the historical information. In this way, voice commands that may have been received from, for example, the display device **208**, an inadvertent activation of the remote control **210**, or the like do not impact the automatic gain processes described herein.

[0080] FIG. **3** illustrates a block diagram of an example media device **306**, according to some embodiments. Media device **306** may include a streaming module **302**, processing module **304**, storage/buffers **308**, and user interface module **307**. As described above, the user interface module **307** may include the audio command processing module **316**.

[0081] The media device **306** may also include one or more audio decoders **312** and one or more video decoders **314**.

[0082] Each audio decoder **312** may be configured to decode audio of one or more audio formats, such as but not limited to AAC, HE-AAC, AC3 (Dolby Digital), EAC3 (Dolby Digital Plus), WMA, WAV, PCM, MP3, OGG GSM, FLAC, AU, AIFF, and/or VOX, to name just some examples.

[0083] Similarly, each video decoder **314** may be configured to decode video of one or more video formats, such as but not limited to MP4 (mp4, m4a, m4v, f4v, f4a, m4b, m4r, f4b, mov), 3GP (3gp, 3gp2, 3g2, 3gpp, 3gpp2), OGG (ogg, oga, ogv, ogx), WMV (wmv, wma, asf), WEBM, FLV, AVI, QuickTime, HDV, MXF (OP1a, OP-Atom), MPEG-TS, MPEG-2 PS, MPEG-2 TS, WAV, Broadcast WAV, LXF, GXF, and/or VOB, to name just some examples. Each video decoder **314** may include one or more video codecs, such as but not limited to H.263, H.264, HEV, MPEG1, MPEG2, MPEG-TS, MPEG-4, Theora, 3GP, DV, DVCPRO, DVCProHD, IMX, XDCAM HD, XDCAM HD422, and/or XDCAM EX, to name just some examples.

[0084] Now referring to both FIGS. **1** and **2**, in some embodiments, the user **232** may interact with the media device **206** via, for example, the remote control **210**. For example, the user **232** may use the remote control **210** to interact with the user interface module **307** of the media device **206** to select content, such as a movie, TV show, music, book, application, game, etc. The streaming module **302** of the media device **206** may request the selected content from the content server(s) **220** over the network **218**. The content server(s) **220** may transmit the requested content to the streaming module **302**. The media device **206** may transmit the received content to the display device **208** for playback to the user **232**.

[0085] In streaming embodiments, the streaming module **302** may transmit the content to the

display device **208** in real time or near real time as it receives such content from the content server(s) **220**. In non-streaming embodiments, the media device **206** may store the content received from content server(s) **220** in storage/buffers **308** for later playback on display device **208**.

[0086] FIG. **4** is a flowchart **400** illustrating example operations for providing a voice command recognition system (VCR), according to some embodiments. Method **400** can be performed by processing logic that can comprise hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), software (e.g., instructions executing on a processing device), or a combination thereof. It is to be appreciated that not all steps may be needed to perform the disclosure provided herein. Further, some of the steps may be performed simultaneously, or in a different order than shown in FIG. **4**, as will be understood by a person of ordinary skill in the art. Without limiting method **400**, method **400** is described with reference to elements of FIG. **1**.

[0087] At **410**, a voice command directed to controlling a device is received, the voice command comprising a wake command and an action command. For example, VCR **102** may receive voice command **104**, including both wake command **114** and action command **116**, via mic **112**.

[0088] At **420**, an amplitude of the wake command is determined. For example, gain processor **118** may measure, calculate, or otherwise identify a wake amplitude **122** corresponding to the wake command **114**. In an embodiment, the wake amp **122** may be identified prior to receiving the action command **116**.

[0089] At **430**, a gain adjustment for the voice command may be calculated based on a comparison of the amplitude of the wake command to a target amplitude. For example, gain processor **118** may calculate gain adjustment **120** based on comparing wake amp **122** to target amp **124**. In an embodiment, the gain adjustment **120** may be mathematical function (addition, subtraction, multiplication) of a number to bring wake amp **122** closer to or within a range of values corresponding to target amp **124**.

[0090] At **440**, an amplitude of the action command is adjusted based on the calculated gain adjustment for the voice command based on the comparison of the amplitude of the wake command to the target amplitude. For example, gain processor **118** may apply gain adjustment **120** to action amp **126** or action command **116** and generate gained action command **128**. In another embodiment, gain processor **118** may also apply a correction **132**, if available.

[0091] At **450**, a device command for controlling the device based on the action command comprising the adjusted amplitude is identified. For example, VCR **102** may identify a device command **108** corresponding to the gained action command **128**. In an embodiment, gained action command **128** may be converted to text, and the text may be used to identify the device command **108**.

[0092] At **460**, the device command is provided to the device. For example, the device command **108** may be “turn on lights” which may be provided to VRD **106**. VRD **106** may be communicatively coupled to living room lights. VRD **106** may then execute the device command **108** communicate to turn on living room lights.

[0093] FIG. **5** is a flowchart **500** illustrating example operations for providing a voice command recognition system (VCR) with beep suppression, according to some embodiments Method **500** can be performed by processing logic that can comprise hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), software (e.g., instructions executing on a processing device), or a combination thereof. It is to be appreciated that not all steps may be needed to perform the disclosure provided herein. Further, some of the steps may be performed simultaneously, or in a different order than shown in FIG. **5**, as will be understood by a person of ordinary skill in the art. Without limiting method **500**, method **500** is described with reference to elements of FIG. **1**.

[0094] At **510**, a wake command for operating a voice responsive device is detected, the voice responsive device being configured to provide an audible confirmation responsive to the detection of the wake command. For example, VCR **102** may receive audio through a mic **112**. From the audio, VCR **102** may detect a wake command **114** associated with operating VRD **106**. The wake

command **114** may be one or more words that signal that an operational or action command **116** is to follow for VRD **106**.

[0095] At **520**, audio received from microphone for one or more sounds. For example, upon detecting wake command **114**, VCR **102** may continue monitoring mic **112** for any immediately subsequent sounds or energy bursts or amplitudes.

[0096] At **530**, one or more sounds received subsequent to the detection of the wake command are determined to indicate a continuous stream of speech. For example, audio receiver **134** may determine that there is no pause, or gap, in speech from user **110** after speaking wake command **114**. In an embodiment, audio receiver **134** may monitor mic **112** for a threshold period of time (e.g., 1 second of $\frac{1}{2}$ second) to determine if additional speech is received from user **110** during the threshold. If additional speech is detected, e.g., based on the amplitude of the received sounds, then audio receiver **134** may determine that a continuous stream of speech is being provided.

[0097] At **540**, the audible confirmation is suppressed responsive to determination of the continuous stream of speech. For example, audio receiver **134** may turn down or turn off the volume on beep **136** which VCR **102** or VRD **106** may be configured to audibly output. In another embodiment, audio receiver **134** may not transmit the beep **136** signal for output upon detecting continuous speech.

[0098] At **550**, an action command issued to the voice responsive device after the wake command is detected from the one or more sounds. For example, VCR **102** may continue to receive and process speech from user **110**, detecting action commands **116** for operating VRD **106**, without the interferences of beep **136**. In an embodiment, the audible beep **136** may be replaced with a visual beep **136**, such as a light, or a text message, or pop up on a screen indicating the wake command **114** has been detected. In another embodiment, the beep **136** may be audibly output upon a detection or completion of receiving action command **116**, in parallel with or prior to executing the action command **116**.

[0099] Various embodiments may be implemented, for example, using one or more well-known computer systems, such as computer system **600** shown in FIG. **6**. For example, the media device **106** may be implemented using combinations or sub-combinations of computer system **600**. Also or alternatively, one or more computer systems **600** may be used, for example, to implement any of the embodiments discussed herein, as well as combinations and sub-combinations thereof.

[0100] Computer system **600** may include one or more processors (also called central processing units, or CPUs), such as a processor **604**. Processor **604** may be connected to a communication infrastructure or bus **606**.

[0101] Computer system **600** may also include user input/output device(s) **603**, such as monitors, keyboards, pointing devices, etc., which may communicate with communication infrastructure **606** through user input/output interface(s) **602**.

[0102] One or more of processors **604** may be a graphics processing unit (GPU). In an embodiment, a GPU may be a processor that is a specialized electronic circuit designed to process mathematically intensive applications. The GPU may have a parallel structure that is efficient for parallel processing of large blocks of data, such as mathematically intensive data common to computer graphics applications, images, videos, etc.

[0103] Computer system **600** may also include a main or primary memory **608**, such as random access memory (RAM). Main memory **608** may include one or more levels of cache. Main memory **608** may have stored therein control logic (i.e., computer software) and/or data.

[0104] Computer system **600** may also include one or more secondary storage devices or memory **610**. Secondary memory **610** may include, for example, a hard disk drive **612** and/or a removable storage device or drive **614**. Removable storage drive **614** may be a floppy disk drive, a magnetic tape drive, a compact disk drive, an optical storage device, tape backup device, and/or any other storage device/drive.

[0105] Removable storage drive **614** may interact with a removable storage unit **618**. Removable

storage unit **618** may include a computer usable or readable storage device having stored thereon computer software (control logic) and/or data. Removable storage unit **618** may be a floppy disk, magnetic tape, compact disk, DVD, optical storage disk, and/or any other computer data storage device. Removable storage drive **614** may read from and/or write to removable storage unit **618**. [0106] Secondary memory **610** may include other means, devices, components, instrumentalities or other approaches for allowing computer programs and/or other instructions and/or data to be accessed by computer system **600**. Such means, devices, components, instrumentalities or other approaches may include, for example, a removable storage unit **622** and an interface **620**. Examples of the removable storage unit **622** and the interface **620** may include a program cartridge and cartridge interface (such as that found in video game devices), a removable memory chip (such as an EPROM or PROM) and associated socket, a memory stick and USB or other port, a memory card and associated memory card slot, and/or any other removable storage unit and associated interface.

[0107] Computer system **600** may further include a communication or network interface **624**. Communication interface **624** may enable computer system **600** to communicate and interact with any combination of external devices, external networks, external entities, etc. (individually and collectively referenced by reference number **628**). For example, communication interface **624** may allow computer system **600** to communicate with external or remote devices **628** over communications path **626**, which may be wired and/or wireless (or a combination thereof), and which may include any combination of LANs, WANs, the Internet, etc. Control logic and/or data may be transmitted to and from computer system **600** via communication path **626**.

[0108] Computer system **600** may also be any of a personal digital assistant (PDA), desktop workstation, laptop or notebook computer, netbook, tablet, smart phone, smart watch or other wearable, appliance, part of the Internet-of-Things, and/or embedded system, to name a few non-limiting examples, or any combination thereof.

[0109] Computer system **600** may be a client or server, accessing or hosting any applications and/or data through any delivery paradigm, including but not limited to remote or distributed cloud computing solutions; local or on-premises software (“on-premise” cloud-based solutions); “as a service” models (e.g., content as a service (CaaS), digital content as a service (DCaaS), software as a service (SaaS), managed software as a service (MSaaS), platform as a service (PaaS), desktop as a service (DaaS), framework as a service (FaaS), backend as a service (BaaS), mobile backend as a service (MBaaS), infrastructure as a service (IaaS), etc.); and/or a hybrid model including any combination of the foregoing examples or other services or delivery paradigms.

[0110] Any applicable data structures, file formats, and schemas in computer system **600** may be derived from standards including but not limited to JavaScript Object Notation (JSON), Extensible Markup Language (XML), Yet Another Markup Language (YAML), Extensible Hypertext Markup Language (XHTML), Wireless Markup Language (WML), MessagePack, XML User Interface Language (XUL), or any other functionally similar representations alone or in combination. Alternatively, proprietary data structures, formats or schemas may be used, either exclusively or in combination with known or open standards.

[0111] In some embodiments, a tangible, non-transitory apparatus or article of manufacture comprising a tangible, non-transitory computer useable or readable medium having control logic (software) stored thereon may also be referred to herein as a computer program product or program storage device. This includes, but is not limited to, computer system **600**, main memory **608**, secondary memory **610**, and removable storage units **618** and **622**, as well as tangible articles of manufacture embodying any combination of the foregoing. Such control logic, when executed by one or more data processing devices (such as computer system **600** or processor(s) **604**), may cause such data processing devices to operate as described herein.

[0112] Based on the teachings contained in this disclosure, it will be apparent to persons skilled in the relevant art(s) how to make and use embodiments of this disclosure using data processing

devices, computer systems and/or computer architectures other than that shown in FIG. 4. In particular, embodiments can operate with software, hardware, and/or operating system implementations other than those described herein.

[0113] It is to be appreciated that the Detailed Description section, and not any other section, is intended to be used to interpret the claims. Other sections can set forth one or more but not all exemplary embodiments as contemplated by the inventor(s), and thus, are not intended to limit this disclosure or the appended claims in any way.

[0114] While this disclosure describes exemplary embodiments for exemplary fields and applications, it should be understood that the disclosure is not limited thereto. Other embodiments and modifications thereto are possible, and are within the scope and spirit of this disclosure. For example, and without limiting the generality of this paragraph, embodiments are not limited to the software, hardware, firmware, and/or entities illustrated in the figures and/or described herein. Further, embodiments (whether or not explicitly described herein) have significant utility to fields and applications beyond the examples described herein.

[0115] Embodiments have been described herein with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined as long as the specified functions and relationships (or equivalents thereof) are appropriately performed. Also, alternative embodiments can perform functional blocks, steps, operations, methods, etc. using orderings different than those described herein.

[0116] References herein to “one embodiment,” “an embodiment,” “an example embodiment,” or similar phrases, indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it would be within the knowledge of persons skilled in the relevant art(s) to incorporate such feature, structure, or characteristic into other embodiments whether or not explicitly mentioned or described herein. Additionally, some embodiments can be described using the expression “coupled” and “connected” along with their derivatives. These terms are not necessarily intended as synonyms for each other. For example, some embodiments can be described using the terms “connected” and/or “coupled” to indicate that two or more elements are in direct physical or electrical contact with each other. The term “coupled,” however, can also mean that two or more elements are not in direct contact with each other, but yet still cooperate or interact with each other.

[0117] The breadth and scope of this disclosure should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. A computer-implemented method, comprising: receiving, by at least one computer processor, a voice command directed to controlling a device, the voice command comprising a wake command and an action command; determining an amplitude of the wake command; determining a previous gain adjustment based on historical information identifying calculated gain adjustments for a plurality of previous voice commands; applying the previous gain adjustment to only the wake command; adjusting, by at least one computer processor, an amplitude of the action command to generate an amplitude-adjusted action command based on at least a difference in the amplitude of the wake command and the amplitude of the action command; identifying a device command for controlling the device based on the amplitude-adjusted action command; and providing the device

command to the device.

2. The computer-implemented method of claim 1, wherein the previous gain adjustment is a median of the calculated gain adjustments for the plurality of previous voice commands.
3. The computer-implemented method of claim 1, further comprising: determining that a calculated gain adjustment of the calculated gain adjustments is an anomaly, wherein the calculated gain adjustment is excluded from the historical information.
4. The computer-implemented method of claim 1, further comprising: identifying a user, from a plurality of users, utilizing a remote control configured to control the device; and determining a time period for evaluating the voice command based on the identified user.
5. The computer-implemented method of claim 1, further comprising: identifying a user, from a plurality of users, utilizing a remote control configured to control the device, wherein the historical information identifies calculated gain adjustments for a plurality of previous voice commands associated with the identified user.
6. The computer-implemented method of claim 1, further comprising: detecting the wake command prior to receiving the action command, wherein the device is configured to output an audible beep upon detecting the wake command; determining that the voice command comprises a continuous stream of speech after detecting the wake command; and suppressing the audible beep based on the continuous stream of speech determination.
7. The computer-implemented method of claim 1, further comprising: retrieving the historical information via a network.
8. A system, comprising: one or more memories; at least one processor each coupled to at least one of the memories and configured to perform operations comprising: receiving a voice command directed to controlling a device, the voice command comprising a wake command and an action command; determining an amplitude of the wake command; determining a previous gain adjustment based on historical information identifying calculated gain adjustments for a plurality of previous voice commands; applying the previous gain adjustment to only the wake command; adjusting an amplitude of the action command to generate an amplitude-adjusted action command based on at least a difference in the amplitude of the wake command and the amplitude of the action command; identifying a device command for controlling the device based on the amplitude-adjusted action command; and providing the device command to the device.
9. The system of claim 8, wherein the previous gain adjustment is a median of the calculated gain adjustments for the plurality of previous voice commands.
10. The system of claim 8, the operations further comprising: determining that a calculated gain adjustment of the calculated gain adjustments is an anomaly, wherein the calculated gain adjustment is excluded from the historical information.
11. The system of claim 8, the operations further comprising: identifying a user, from a plurality of users, utilizing a remote control configured to control the device; and determining a time period for evaluating the voice command based on the identified user.
12. The system of claim 8, the operations further comprising: identifying a user, from a plurality of users, utilizing a remote control configured to control the device, wherein the historical information identifies calculated gain adjustments for a plurality of previous voice commands associated with the identified user.
13. The system of claim 8, the operations further comprising: detecting the wake command prior to receiving the action command, wherein the device is configured to output an audible beep upon detecting the wake command; determining that the voice command comprises a continuous stream of speech after detecting the wake command; and suppressing the audible beep based on the continuous stream of speech determination.
14. The system of claim 8, the operations further comprising: retrieving the historical information via a network.
15. A non-transitory computer-readable medium having instructions stored thereon that, when

executed by at least one computing device, cause the at least one computing device to perform operations comprising: receiving a voice command directed to controlling a device, the voice command comprising a wake command and an action command; determining an amplitude of the wake command; determining a previous gain adjustment based on historical information identifying calculated gain adjustments for a plurality of previous voice commands; applying the previous gain adjustment to only the wake command; adjusting an amplitude of the action command to generate an amplitude-adjusted action command based on at least a difference in the amplitude of the wake command and the amplitude of the action command; identifying a device command for controlling the device based on the amplitude-adjusted action command; and providing the device command to the device.

16. The non-transitory computer-readable medium of claim 15, wherein the previous gain adjustment is a median of the calculated gain adjustments for the plurality of previous voice commands.

17. The non-transitory computer-readable medium of claim 15, the operations further comprising: determining that a calculated gain adjustment of the calculated gain adjustments is an anomaly, wherein the calculated gain adjustment is excluded from the historical information.

18. The non-transitory computer-readable medium of claim 15, the operations further comprising: identifying a user, from a plurality of users, utilizing a remote control configured to control the device; and determining a time period for evaluating the voice command based on the identified user.

19. The non-transitory computer-readable medium of claim 15, the operations further comprising: identifying a user, from a plurality of users, utilizing a remote control configured to control the device, wherein the historical information identifies calculated gain adjustments for a plurality of previous voice commands associated with the identified user.

20. The non-transitory computer-readable medium of claim 15, the operations further comprising: detecting the wake command prior to receiving the action command, wherein the device is configured to output an audible beep upon detecting the wake command; determining that the voice command comprises a continuous stream of speech after detecting the wake command; and suppressing the audible beep based on the continuous stream of speech determination.
